Cardiovascular System (CARD) · NPLEX-P1-CARD-18

# CV-18-I Vascular Conditions

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

| Competency | BD1 done | BD2 done | Confidence 1 to 4 | Next review date |
|---|---|---|---|---|
| NPLEX-P1-CARD-18-09-C01 | | | | |
| NPLEX-P1-CARD-18-09-C02 | | | | |
| NPLEX-P1-CARD-18-09-C03 | | | | |
| NPLEX-P1-CARD-18-09-C04 | | | | |
| NPLEX-P1-CARD-18-09-C05 | | | | |
| NPLEX-P1-CARD-18-09-C06 | | | | |
| NPLEX-P1-CARD-18-09-C07 | | | | |
| NPLEX-P1-CARD-18-09-C08 | | | | |
| NPLEX-P1-CARD-18-09-C09 | | | | |
| NPLEX-P1-CARD-18-09-C10 | | | | |
| NPLEX-P1-CARD-18-09-C11 | | | | |
| NPLEX-P1-CARD-18-09-C12 | | | | |
| NPLEX-P1-CARD-18-09-C13 | | | | |
| NPLEX-P1-CARD-18-09-C14 | | | | |
| NPLEX-P1-CARD-18-09-C15 | | | | |
| NPLEX-P1-CARD-18-09-C16 | | | | |
| NPLEX-P1-CARD-18-09-C17 | | | | |

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C01-BD1

# Rebuild the three patterns of arteriosclerosis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C01 Activity: NPLEX-P1-CARD-18-09-C01-BD1 Date: _____

Without looking at your notes, write out the three patterns of arteriosclerosis. For each one write the size and type of vessel involved, the layer of the wall that is damaged, what the lesion looks like down the microscope, and whether the lumen narrows. Under arteriolosclerosis, separate the hyaline form from the hyperplastic form and write the clinical setting each belongs to, what the kidney looks like in each, and which one shows fibrinoid necrosis. Finish with one sentence on why the term arteriosclerosis is not the same as atherosclerosis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C01-BD2

## Two patients with hardened arteries

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C01 Activity: NPLEX-P1-CARD-18-09-C01-BD2 Date: _____

A 72-year-old man with type 2 diabetes and chronic kidney disease has hard, cord-like leg arteries, pipestem calcification on a foot radiograph, and an ankle-brachial index of 1.52. A 44-year-old woman with a blood pressure of 220/128 mm Hg has pinpoint hemorrhages on the renal cortical surface and arterioles with concentric layers of smooth muscle. For each patient write which pattern of arteriosclerosis is present, which layer of the wall holds the lesion, and whether flow through the vessel is reduced. For the first patient write why the ankle pressure reads high and what that does to the usefulness of the index. For the second patient write what the layered appearance is made of and why the kidney becomes ischemic. Finish by writing what a kidney with a finely granular cortical surface and glassy pink arterioles tells you instead.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table  NPLEX-P1-CARD-18-09-C01-CP01

## The patterns of arterial hardening side by side

Fill in every cell from memory, then open the notes and correct anything you missed.

| Feature | Atherosclerosis | Hyaline arteriolosclerosis | Hyperplastic arteriolosclerosis | Monckeberg medial calcific sclerosis |
|---|---|---|---|---|
| Vessels involved | | | | |
| Layer of the wall affected | | | | |
| What the lesion is made of | | | | |
| Effect on the lumen and on flow | | | | |
| Typical clinical setting | | | | |
| Effect on a measured ankle pressure | | | | |

Confidence before checking

☐ 1  Guessing  ☐ 2  Developing  ☐ 3  Nearly competent  ☐ 4  Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C02-BD1

# Build an atheroma from the first injury

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C02 Activity: NPLEX-P1-CARD-18-09-C02-BD1 Date: _____

With your notes closed, write the sequence that turns a normal artery into a complicated plaque, one step per line and in order, from the first endothelial change to the final events that make a plaque dangerous. Beside the step where macrophages load with lipid, name the receptor they use and write why that uptake does not switch off. Then draw a cross section of a mature atheroma and label the fibrous cap and the core, writing what each is made of and which cell builds the cap. List the arteries that plaque favors, in order from the most affected, and separate the modifiable risk factors from the fixed ones.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C02-BD2

## Why a small plaque can kill and a big one can ache

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C02 Activity: NPLEX-P1-CARD-18-09-C02-BD2 Date: _____

Patient A has a 40% stenosis of the left anterior descending artery and dies of an infarction in that vessel. Patient B has an 80% stenosis of the same artery and has chest pain only when climbing stairs. Explain each outcome from the structure of the plaque rather than its size. Write the three features of a plaque that make it likely to rupture, and name the cell and the enzyme that thin the cap along with the cytokine that stops smooth muscle replacing the collagen. Write what is exposed when the cap tears and what forms on top of it. Then write the degree of stenosis at which flow can no longer rise with demand, and explain what that means for when patient B feels symptoms. Finish by listing the five complications of a plaque with the mechanism of each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-09-C02-RT01

## Rebuild the atherogenesis sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C02 Activity: NPLEX-P1-CARD-18-09-C02-RT01 Date: _____

Rebuild the eight-step sequence from endothelial injury to complicated plaque, one step per line and in order, with the steps numbered. Beside the first step, write where in the arterial tree the injury starts and why that location and not another. Beside the step where monocytes enter, name the adhesion molecule the endothelium displays. Beside the step where smooth muscle arrives, name the growth factor that recruits it and the layer it comes from. Mark on your list the first lesion that is visible to the naked eye and write at what age it is already present in most people, and mark the first step at which the lumen is meaningfully narrowed.

Rebuild from memory: From endothelial injury to complicated plaque

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question  NPLEX-P1-CARD-18-09-C02-Q01

## An infarct from a modest stenosis

**A 57-year-old man dies suddenly at home. An angiogram performed four months earlier had shown a 40% stenosis of the proximal left anterior descending artery and no other significant lesions. At autopsy the left anterior descending artery is occluded by a fresh thrombus overlying a plaque with a thin fibrous cap, a large lipid core, and dense macrophage infiltration. Which of the following best explains why this plaque occluded the artery?**

A. Calcification of the media stiffened the artery and reduced its luminal diameter

B. Gradual growth of the plaque reduced coronary flow reserve beyond the ischemic threshold

C. Macrophage metalloproteinases degraded cap collagen while interferon gamma suppressed its replacement

D. Plasma proteins leaked into the wall and smooth muscle cells laid down excess matrix

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C03-BD1

## Draw LDL clearance and then break it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C03 Activity: NPLEX-P1-CARD-18-09-C03-BD1 Date: _____

Draw a hepatocyte and trace one LDL particle from plasma into the lysosome. Label the apolipoprotein the receptor binds, the mechanism of uptake, and what the released cholesterol does to SREBP-2, to HMG-CoA reductase, and to the number of new receptors. Then write the three genes that cause familial hypercholesterolemia, stating for each whether the mutation is loss of function or gain of function and which part of the pathway it breaks. Write the inheritance pattern. Finish by listing the physical deposits of this disease, writing for each where it sits, what it is made of, and which ones are specific enough to point at the diagnosis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C03-BD2

## Read the lipid panel and the tendons

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C03 Activity: NPLEX-P1-CARD-18-09-C03-BD2 Date: _____

A 9-year-old boy has an LDL cholesterol of 620 mg/dL, yellow plaques over the knuckles and elbows, and a father who had a myocardial infarction at 41. His 38-year-old mother has an LDL of 236 mg/dL, normal triglycerides, a firm nodule in the Achilles tendon, and a gray-white ring at the corneal margin. For each of them write the likely receptor activity, whether they are heterozygous or homozygous, and the decade in which coronary disease is expected. Explain why the plasma LDL rises even though the liver is detecting too little cholesterol inside its own cells. Write why normal triglycerides matter for this diagnosis. Then write why the corneal ring is strong evidence in the mother but would be weak evidence in a 70-year-old, and explain in two or three sentences how the same receptor failure that raises plasma LDL also builds the tendon nodule and the arterial plaque.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C04-BD1

## Build the aneurysm table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C04 Activity: NPLEX-P1-CARD-18-09-C04-BD1 Date: _____

On blank paper build a table of four aneurysms, one row each for abdominal aortic, thoracic aortic, berry, and mycotic. Use the columns site, what weakens the wall, typical patient or association, and main consequence, and fill every cell. Above the table write the definition of an aneurysm, the difference between a true and a false aneurysm, and the diameter at which an abdominal aorta is called aneurysmal. Below the table write the formula that relates wall stress to pressure, radius, and wall thickness, and write two sentences on why dilation tends to continue once it has started. Finish by writing why the infrarenal aorta is the usual site, naming what supplies that part of the media and what plaque does to it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C04-BD2

## Work the numbers on a growing aorta

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C04 Activity: NPLEX-P1-CARD-18-09-C04-BD2 Date: _____

A 69-year-old man who smoked for 40 years has an infrarenal aorta measuring 6.0 cm. A normal aorta at that level measures 2.0 cm. Calculate how many times greater the wall stress is in his aorta at the same blood pressure, show your working, and write what happens to the stress as the stretched wall thins. Explain why his rupture risk is far higher than that of a man with a 4.0 cm aorta, and write what uncontrolled hypertension adds to every step of your reasoning. Then write the three findings expected if this aneurysm ruptures. Separately, a 24-year-old tall man with long fingers has a dilated ascending aorta and a new diastolic murmur: write which layer has degenerated, what that degeneration looks like microscopically, and why the valve leaks when the root widens. Finish by writing how tertiary syphilis produces the same root dilation through a different route.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C05-BD1

## Rebuild a dissection and its consequences

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C05 Activity: NPLEX-P1-CARD-18-09-C05-BD1 Date: _____

Write the definition of aortic dissection, naming where the blood enters, which layer it travels in, and what the resulting channel is called. List the conditions that weaken the media, separating the common cause in older patients from the inherited and structural causes seen in younger patients, and name the acute triggers. Then rebuild the seven-step sequence of a proximal dissection from weakened media to external rupture, one step per line, and mark which step is the usual cause of death. Draw the Stanford classification as two labelled diagrams and write beside each what it does and does not involve. Finish by writing why heavy atherosclerosis makes a dissection less likely to spread.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C05-BD2

## Explain each sign in a dissecting aorta

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C05 Activity: NPLEX-P1-CARD-18-09-C05-BD2 Date: _____

A 58-year-old man with long-standing hypertension has sudden tearing chest pain that moves to between his shoulder blades. Blood pressure is 176/96 mm Hg in the right arm and 128/74 mm Hg in the left, there is a new diastolic murmur, and the chest radiograph shows a widened mediastinum. Explain each of those four findings from the anatomy of the false lumen, writing one short paragraph per finding. Then he becomes hypotensive with distended neck veins and muffled heart sounds: write where the aorta has ruptured, why filling of the ventricles falls, and which Stanford type this must be. Write the Stanford type you would expect instead if the tear had begun beyond the left subclavian artery, and list the complications that type threatens. Finish by writing which single finding in this man would point you to branch occlusion of the renal, mesenteric, carotid, or spinal arteries.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question  NPLEX-P1-CARD-18-09-C05-Q02

## Tearing chest pain in a tall young man

**A 31-year-old man who is 196 cm tall with long slender fingers and a high arched palate comes to the emergency department because of sudden severe tearing chest pain radiating to his back. Blood pressure is 88/54 mm Hg, the jugular veins are distended, and the heart sounds are muffled. Which of the following best explains his hypotension?**

A. Blood has ruptured from the aorta into the pericardial sac and is limiting ventricular filling

B. The false lumen has occluded both renal arteries and reduced circulating volume

C. Fibrinoid necrosis of the arterioles has produced widespread loss of vascular tone

D. An intimal plaque has starved the underlying media and allowed the aorta to dilate

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C06-BD1

## Claudication, the ankle-brachial index, and the cholesterol shower

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C06 Activity: NPLEX-P1-CARD-18-09-C06-BD1 Date: _____

Write what peripheral arterial disease is and name its two strongest risk factors. Explain in three or four sentences why the leg hurts with walking and not at rest, naming what the arterioles do during exercise and why the stenosis prevents flow from rising. Write which muscle group hurts with superficial femoral disease and which hurts with aortoiliac disease. Then rebuild the ankle-brachial index table from memory with a row for each band, giving the numeric range and what it means, and write the one band in which the result cannot be trusted and why. List the findings of chronic limb ischemia on inspection and palpation, write what rest pain means and why hanging the foot down eases it, and name the three features of Leriche syndrome. Finish with atheroembolic disease: write what showers downstream, what procedures set it off, the size of vessel the crystals lodge in, the skin and kidney findings, why the pedal pulses stay normal, and what the biopsy shows.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C06-BD2

## Three painful feet, three different vessels

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C06 Activity: NPLEX-P1-CARD-18-09-C06-BD2 Date: _____

Patient 1 is a 70-year-old smoker with calf cramping after one block of walking that stops within three minutes of standing still, cool hairless shins, absent pedal pulses, and an ankle-brachial index of 0.58. Patient 2 is a 66-year-old who had coronary angiography four days ago and now has purple toes, a mottled net-like pattern on the thighs, rising creatinine, eosinophilia, and normal pedal pulses. Patient 3 is a 74-year-old with diabetes on dialysis who has a nonhealing great toe ulcer and an ankle-brachial index of 1.46. For each patient name the vascular process, write which size of vessel is obstructed and by what material, and explain how the pulse finding follows from that. Explain why patient 2 keeps normal pulses while patient 1 loses them. Write why patient 3's index does not exclude disease and what you would measure instead to judge perfusion. Finish by writing which patient has an ulcer you would expect to be punched out and dry rather than shallow and moist, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise  NPLEX-P1-CARD-18-09-C06-PR01

## Predict what happens as a femoral stenosis tightens

A 68-year-old smoker has a single stenosis of the superficial femoral artery that narrows steadily over three years. For each item below write the direction of the change and give the one-line mechanism that explains it.

| What changes | Up, down, or no change | Why, in one or two lines |
|---|---|---|
| What happens to resting blood flow in the calf while the stenosis is still moderate | | |
| What happens to calf blood flow during walking, and what the patient feels | | |
| What happens to the ankle systolic pressure and therefore to the ankle-brachial index | | |
| What happens to the distance he can walk before the pain starts | | |
| What happens to pain at rest in the forefoot at night once the stenosis is severe, and what hanging the foot over the edge of the bed does to it | | |
| What happens to the skin, nails, and hair of the lower leg, and what appears at the toes and pressure points | | |

Confidence before checking

☐ 1  Guessing  ☐ 2  Developing  ☐ 3  Nearly competent  ☐ 4  Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C07-BD1

## Rebuild pulmonary embolism from clot to blood gas

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C07 Activity: NPLEX-P1-CARD-18-09-C07-BD1 Date: _____

Write where more than 95% of pulmonary emboli come from, name the specific veins, and write why those veins and not the calf veins alone. Then rebuild the seven-step sequence from arterial obstruction to respiratory alkalosis, one step per line and in order, and write beside the fourth step what is happening to ventilation and perfusion in the obstructed region and in the open region. Write the proportion of the pulmonary circulation whose obstruction can cause shock or sudden death. List the clinical findings of a large embolus and separately those that suggest a small peripheral one. Finish by writing the findings of acute right ventricular strain, including the most common electrocardiographic finding, and write what D-dimer is and the one limitation of the test.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C07-BD2

## Calculate the gradient and explain the lung that lives

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C07 Activity: NPLEX-P1-CARD-18-09-C07-BD2 Date: _____

A 29-year-old woman taking an estrogen-containing contraceptive breathes room air at sea level. Her PaO2 is 58 mm Hg and her PaCO2 is 28 mm Hg. Calculate the alveolar PO2 and the alveolar-arterial gradient, showing every step of the arithmetic, and state whether the gradient is normal for a young adult. Write why her low PaCO2 rules out hypoventilation as the cause of her hypoxemia, and name the mechanism that explains it instead. Then write why only about one in ten emboli infarcts the lung, naming the two things that keep the tissue alive, and write which coexisting condition makes infarction likely and why. Describe what the infarct looks like and where it sits. Finish by writing what repeated small emboli over years produce.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-09-C07-RT02

## Rebuild the pulmonary embolism sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C07 Activity: NPLEX-P1-CARD-18-09-C07-RT02 Date: _____

Rebuild the seven-step sequence from pulmonary arterial obstruction to respiratory alkalosis, numbering each step and writing one step per line. Beside the step where pulmonary vascular resistance rises, write why the right ventricle is the chamber that fails and what that failure is called. Beside the step where blood is diverted, draw two alveolar units side by side, one obstructed and one open, and mark on each what is happening to ventilation and to perfusion. Beside the hypoxemia step, write which measurement widens and what that measurement separates this condition from. Finish by writing which step produces the low $PaCO_2$ and why the patient is breathing faster.

Rebuild from memory: From pulmonary arterial obstruction to signs

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-09-C07-CA01

## Sudden breathlessness after a long flight

A 34-year-old woman comes to the emergency department because of 2 hours of sudden breathlessness and sharp right-sided chest pain that is worse on deep inspiration. She returned yesterday from a 14-hour flight and has been taking a combined oral contraceptive for six years. Her left calf has been swollen and tender for five days. Temperature is 37.4 degrees C (99.3 degrees F), pulse is 116/min, respirations are 28/min, blood pressure is 104/68 mm Hg, and oxygen saturation is 89% on room air. The jugular venous pressure is raised and there is a loud pulmonary component of the second heart sound. Arterial blood gas on room air at sea level shows PaO2 56 mm Hg and PaCO2 29 mm Hg. The electrocardiogram shows sinus tachycardia. The chest radiograph is clear.

1. Recognize. Name the vascular event, the vein the material most likely came from, and the three elements of risk present in this woman.

2. Explain. Explain her hypoxemia and her low PaCO2 mechanistically, naming what is happening to ventilation and perfusion in the obstructed and the open lung regions.

3. Differentiate. Write how the blood gas pattern here differs from that of pure hypoventilation and from that of a right-to-left shunt, naming the measurement that separates them.

4. Investigate. Calculate the alveolar-arterial gradient from the values given, show the arithmetic, and state what the result tells you; then name the leg finding you would look for on compression ultrasound and what it would show.

5. Decide. State which chamber of the heart is under acute strain and name the three findings in this case that point to it.

6. Justify. Justify from the dual blood supply of the lung why her lung tissue is likely to survive, and write what would change that conclusion.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C08-BD1

## Build Virchow triad and the venous thrombus

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C08 Activity: NPLEX-P1-CARD-18-09-C08-BD1 Date: _____

Write where venous thrombi usually begin and why that exact site, and write what they are made of and how that differs from an arterial thrombus. Then build a table of Virchow triad with one row per element, giving the mechanism and at least four examples for each, and mark the two clinical situations that supply all three elements at once. Write the inherited and acquired causes of hypercoagulability from the notes, including the one that works by losing a protein in the urine. Finish by writing what a deep vein thrombosis does to capillary hydrostatic pressure and what the leg looks and feels like, why many are silent, which clinical sign is unreliable, and what the thrombosed vein does under the ultrasound probe.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C08-BD2

## A red cord, a swollen leg, and a wandering vein

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C08 Activity: NPLEX-P1-CARD-18-09-C08-BD2 Date: _____

Patient 1 has a tender red palpable cord along a varicose vein on the calf with no limb swelling. Patient 2 has a uniformly swollen, warm, tender left leg with a calf circumference 4 cm greater than the right and dilated superficial veins. Patient 3 is a 61-year-old with 48 kg weight loss and painless jaundice who has had thrombophlebitis in the left arm, then the right calf, then the left thigh over ten weeks. For each patient name the venous condition, write which venous system is involved, and write the embolic risk with the reason. Explain why patient 2 has diffuse swelling and patient 1 does not. For patient 3 name the sign, the tumor type most often responsible, the organ to suspect first, and the substances the tumor releases that activate coagulation, then write which identically named sign belongs to a completely different disorder and what that other sign actually is. Finish by writing which of these three patients you would expect to have a normal platelet count and why that question is not useful here.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question  NPLEX-P1-CARD-18-09-C08-Q03

## Thrombosis at shifting sites

**A 63-year-old woman has had 14 kg of weight loss, painless jaundice, and three episodes of tender palpable venous cords over ten weeks, first in the right forearm, then the left calf, then the right thigh, each resolving as the next appeared. Her platelet count and coagulation studies are normal. Which of the following best explains her venous findings?**

A. Antiphospholipid antibodies binding phospholipid-protein complexes on the endothelium

B. Loss of antithrombin in the urine from nephrotic syndrome

C. Compression of the deep veins by an enlarging pelvic mass

D. Tissue factor and mucin released by an adenocarcinoma activating coagulation

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C09-BD1

## The calf pump, the valves, and the skin that follows

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C09 Activity: NPLEX-P1-CARD-18-09-C09-BD1 Date: _____

Draw a leg vein in outline with a valve, a perforating vein, and the calf muscle. Write how blood is returned against gravity during walking, naming the pump and the direction competent valves allow through the perforators, and write what happens to ankle venous pressure during movement when the valves work. Then write what happens to that pressure when the valves fail and name the resulting state. Define varicose veins, name the venous system most often involved, and list the risk factors with a one-line reason for pregnancy and for prolonged standing. Write which other sites develop varicosities when pressure rises elsewhere in the body and what raises it. Finish by rebuilding the seven-step sequence from valve incompetence to a venous ulcer and write beside the pigment step what the brown colour is made of.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C09-BD2

## Two ulcers above two different problems

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C09 Activity: NPLEX-P1-CARD-18-09-C09-BD2 Date: _____

A 64-year-old woman who stands all day at work has ankle swelling that is worst in the evening, brown discoloration and itchy scaling skin around both lower legs, indurated tissue above the ankles, and a shallow moist irregular ulcer just above the left medial malleolus. A 71-year-old man who smokes has a dry, pale, punched out ulcer on the tip of the second toe, a cool hairless foot, and absent pedal pulses. For each ulcer write the underlying vascular problem, the usual location and why that location, the appearance, the character of the pain, and the state of the surrounding skin. Explain why elevating the leg helps one patient and harms the other, giving the pressure that changes in each case. Then explain each of the woman's four skin findings from the steps of the venous sequence, one short paragraph each. Finish by writing why her swelling is worst in the evening and better by morning.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-09-C09-RT03

## Rebuild the path from a failed valve to a broken skin surface

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C09 Activity: NPLEX-P1-CARD-18-09-C09-RT03 Date: _____

Rebuild the seven-step sequence from valve incompetence to a venous ulcer, numbering each step and writing one step per line. Beside the first step, write the three groups of veins in which the valves can fail and what a previous deep vein thrombosis does to deep valves. Beside the edema step, write which pressure has risen and in which vessel. Beside the pigment step, write what leaks out of the capillary and what the brown pigment is. Beside the oxygenation step, write the two changes that impair delivery of oxygen to the skin. Then mark on the list the step at which the changes become visible as a skin disease and name that disease and the fibrotic stage that follows.

Rebuild from memory: From incompetent valves to stasis ulcer

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval  NPLEX-P1-CARD-18-09-C10-BD1

## The three colours of a vasospastic attack

NPLEX Part: Part I  Subject: Cardiovascular System (CARD)  Chapter and topic: CV-18-I Vascular Conditions  Competency: NPLEX-P1-CARD-18-09-C10  Activity: NPLEX-P1-CARD-18-09-C10-BD1  Date: _____

Write the definition of Raynaud phenomenon, naming the vessels involved, the body parts affected, and the two triggers. Write how digital vessels normally respond to cold, naming the nerve supply and the specific receptor subtype, and write the purpose of that response. Then draw the attack as a sequence of five steps and label the three colours in order, writing beside each colour what the blood in the digit is doing to produce it. Explain the red phase in full, naming at least three local metabolites that accumulate during ischemia and what they do to the arterioles. Finish by writing which colour is the most specific and why not every attack shows all three.

My response

Confidence before checking

☐ 1  Guessing  ☐ 2  Developing  ☐ 3  Nearly competent  ☐ 4  Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C10-BD2

## Which Raynaud is in front of you

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C10 Activity: NPLEX-P1-CARD-18-09-C10-BD2 Date: _____

Patient 1 is a 19-year-old woman whose fingers of both hands turn white then blue in cold weather, with no ulcers, normal nailfold capillaries, and a negative antinuclear antibody. Patient 2 is a 47-year-old woman with attacks confined to three fingers of the right hand, pitted scars on two fingertips, tight shiny skin over the fingers, and dilated and dropped-out nailfold capillaries. For each patient write whether the phenomenon is primary or secondary, what is wrong with the vessel, and whether tissue damage is expected. Explain why one patient can lose tissue and the other cannot, using the state of the lumen between attacks. Name the disease most likely behind patient 2 and write what it does to the digital arteries. Then list the other conditions and exposures that cause secondary disease. Finish by writing the five features that should make you look for an underlying disease in any patient with cold-triggered colour change.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C11-BD1

## Thromboangiitis obliterans from histology to toes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C11 Activity: NPLEX-P1-CARD-18-09-C11-BD1 Date: _____

Write the definition of thromboangiitis obliterans, naming the vessels involved, the arteries most often affected, and the two other structures the inflammation spreads to. Write who gets it, the age at onset, and the single exposure that is central to it, with what is thought to be the mechanism and what happens to the disease when that exposure stops. Then describe the microscopic lesion in full, naming what is inside the thrombus, what surrounds those collections, what happens to the internal elastic lamina, and what the segment looks like over time. Write what the word segmental means here and why it separates this disease from atherosclerosis. Finish by listing the clinical features in order from earliest to latest, writing beside the claudication which part of the limb hurts and why, and beside the rest pain why it is out of proportion to the visible lesion.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C11-BD2

## A 33-year-old smoker with a black fingertip

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C11 Activity: NPLEX-P1-CARD-18-09-C11-BD2 Date: _____

A 33-year-old man who smokes 30 cigarettes a day has pain in the instep of the right foot when walking, a painful ulcer on the tip of the left index finger, absent radial and posterior tibial pulses with normal brachial and femoral pulses, cold-triggered colour change in both hands, and a tender nodular cord that appeared in one calf vein and then another. Write the diagnosis and then justify it finding by finding, writing one line per finding. Explain why the proximal pulses are normal while the distal ones are absent. Write why an upper limb finding argues against atherosclerotic disease in a man of this age, and write three further features that separate the two conditions. Explain what the vein nodules and the colour change tell you about the extent of the disease beyond the arteries. Finish by writing what determines whether this man keeps his fingers and toes, naming the mechanism behind that answer.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C12-BD1

## Sort the vasculitides by vessel size, then build giant cell arteritis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C12 Activity: NPLEX-P1-CARD-18-09-C12-BD1 Date: _____

Write the definition of vasculitis and the principle on which the Chapel Hill classification is built. For each of the three vessel sizes write what the clinical picture looks like and why. Then rebuild the vasculitis table from memory with a row for each disease, using the columns vessel size, main immune mechanism, and key association, and include the large, medium, small ANCA, small immune complex, and variable vessel diseases. Next write giant cell arteritis in full: the age at which it starts, the sex and ancestry most affected, the immune mechanism, and every artery named in the notes. Write the cytokine that drives the acute-phase response, then the three ischemic symptoms with the mechanism of each, the laboratory findings with values, and what the biopsy shows. Write why a biopsy can be negative in a patient who has the disease. Finish with polymyalgia rheumatica: the girdles involved, the laboratory result, why strength and creatine kinase are normal, and how often the two conditions overlap.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C12-BD2

## Jaw pain, a headache, and sudden loss of vision

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C12 Activity: NPLEX-P1-CARD-18-09-C12-BD2 Date: _____

A 74-year-old woman of Northern European ancestry has six weeks of new temporal headache with a tender thickened cord over the temple, pain in the jaw after a few minutes of chewing, aching and morning stiffness of both shoulders and hips, and this morning sudden painless loss of vision in the left eye. The erythrocyte sedimentation rate is 86 mm/h, C-reactive protein is high, creatine kinase is normal, and shoulder strength is normal. Explain each of her five problems mechanistically, one short paragraph each, naming the artery involved where there is one. Write why her shoulder and hip symptoms are not a muscle disease and what tissue is actually inflamed. Write why the sedimentation rate is so high, naming the cytokine responsible. Then a temporal artery biopsy is reported as normal: write what that does and does not tell you and the feature of the lesion that explains it. Finish by writing which vessel this disease can involve that would later produce a thoracic aneurysm.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C13-BD1

## Takayasu arteritis from arch to pulse

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C13 Activity: NPLEX-P1-CARD-18-09-C13-BD1 Date: _____

Write what Takayasu arteritis is, naming every vessel group the notes list, the usual age and sex, and the region where it is more common. Describe the histology and write which other vasculitis it resembles and the single feature that separates the two. Then write what chronic inflammation leaves behind in the media, adventitia, and intima, and what that does to the origins of the great vessels. Write the constitutional symptoms that come first and the laboratory finding that accompanies them. Explain why the arm pulses weaken, name the older name of the disease, and write what happens to the blood pressure in the affected arm and what is heard over the subclavian and carotid arteries. Finish by writing the findings produced by carotid and vertebral narrowing, by renal artery stenosis, by aortic root involvement, and by pulmonary artery involvement.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C13-BD2

## Two women with a weak arm pulse

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C13 Activity: NPLEX-P1-CARD-18-09-C13-BD2 Date: _____

A 28-year-old woman has three months of fever, fatigue, and weight loss, then arm aching with use, a blood pressure of 94/62 mm Hg in the left arm and 148/92 mm Hg in the right, a bruit over the left subclavian artery, and a sedimentation rate of 72 mm/h. A 61-year-old man with hypertension has sudden tearing chest pain radiating between his shoulder blades and a blood pressure of 182/98 mm Hg in the right arm and 126/76 mm Hg in the left. For each patient write the diagnosis and the mechanism by which the arm pressures became unequal. Write which features of the history separate them, and explain why the time course is the most useful single discriminator. For the woman, write why a blood pressure taken in her left arm would underestimate her true pressure and what that means if she also has renal artery stenosis. Then write what you would expect her blood pressure to do if the renal arteries are involved and name the system that produces it. Finish by writing which disease a granulomatous large-vessel arteritis would be in a 68-year-old instead, and the single feature that decides it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table  NPLEX-P1-CARD-18-09-C13-CP02

## The two granulomatous large-vessel arteritides

Fill in every cell from memory, then open the notes and correct anything you missed.

| Feature | Giant cell arteritis | Takayasu arteritis |
|---|---|---|
| Age at onset | | |
| Typical patient and ancestry | | |
| Main vessels involved | | |
| Key clinical findings | | |
| Histology | | |
| Laboratory findings | | |
| Late aortic complication | | |

Confidence before checking

☐ 1  Guessing  ☐ 2  Developing  ☐ 3  Nearly competent  ☐ 4  Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C14-BD1

## Polyarteritis nodosa, vessel by vessel

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C14 Activity: NPLEX-P1-CARD-18-09-C14-BD1 Date: _____

Write the definition of polyarteritis nodosa, naming the size and type of vessel involved, the two organ groups most often affected, and the three vessel types that are spared. Write what that sparing means for the kidney and name the lesion the disease therefore cannot cause. Write the age group affected, the infection associated with a substantial minority of cases, the mechanism in those cases including the hypersensitivity type and the antigen involved, and the antibody result that is characteristically negative. Then describe the lesion: what is happening across the wall, the cells present early, what the necrosis does to the wall, and what that produces on angiography and in the name of the disease. Write the hallmark that concerns the age of the lesions. Finish by listing the organ manifestations, giving the mechanism of each, and name the leading cause of death.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C14-BD2

## Explain a patient with fever, foot drop, and bloody stools

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C14 Activity: NPLEX-P1-CARD-18-09-C14-BD2 Date: _____

A 46-year-old man with chronic hepatitis B has eight weeks of fever, malaise, and 9 kg weight loss, new hypertension at 168/104 mm Hg, abdominal pain that begins 30 minutes after eating with dark stools, a right foot that will not dorsiflex, painful nodules on the calves, and a creatinine of 2.1 mg/dL with no red cell casts and no hematuria. Chest imaging is normal and ANCA is negative. Write the diagnosis and then explain each finding mechanistically, one short paragraph per finding, naming the vessel involved. Explain why his urine shows no red cell casts and what that tells you about the size of vessel affected. Write why the hepatitis B infection matters and what would be found in his vessel walls. Write what renal and mesenteric angiography would show and why the wall produces that appearance. Then write which small-vessel vasculitis you would have suspected instead if he had hemoptysis, red cell casts, and a positive ANCA. Finish by writing what the coexistence of acute and healed lesions in the same artery tells you about how the disease injures vessels.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C15-BD1

## Kawasaki disease and the coronary arteries

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C15 Activity: NPLEX-P1-CARD-18-09-C15-BD1 Date: _____

Write what Kawasaki disease is, naming the size of vessel involved, the age group and ancestry most affected, and why it matters in children as a group. Write the current view of what triggers it and which cells damage the vessel wall, then describe the arteritis and write how it differs in one respect from the lesion of polyarteritis nodosa. Write the fever requirement for the clinical diagnosis and list the five other principal features with a short description of each, then write how many of them are needed alongside the fever. Write what happens to the fingers and toes in the second to third week and list the laboratory findings, including the one that appears only after the first week. Finish by writing the key complication, how often it occurs in untreated children, the mechanism that produces it, the two things inside an aneurysm that favour clot, what that clot can cause, and the other cardiac problems of the acute phase.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C15-BD2

## A red-eyed, red-tongued, miserable toddler

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C15 Activity: NPLEX-P1-CARD-18-09-C15-BD2 Date: _____

A 3-year-old boy has had a temperature of 39.4 degrees C (103 degrees F) for six days, red eyes without discharge, cracked red lips with a strawberry tongue, a rash on the trunk and perineum, swollen red hands and feet, and one enlarged cervical lymph node measuring 2 cm. He is extremely irritable. Laboratory testing shows a raised sedimentation rate and C-reactive protein, leukocytosis, and a platelet count of 610,000/microL. Write the diagnosis and then write which features you are counting towards the clinical criteria and whether the criteria are met. Write what the eye finding is called and the detail that distinguishes it, and write what you expect his fingers to do next week. Explain the high platelet count and when it appears. Then write which vessels you are now worried about, the mechanism by which they dilate, and why thrombosis is likely inside the resulting lesion, and name what that thrombosis can cause in a child of this age. Finish by writing why scarlet fever does not explain this picture, naming the organism, the two features of this boy that scarlet fever would not produce, and the complication it does not cause.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C16-BD1

## How ANCA injures a capillary, and the three diseases that follow

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C16 Activity: NPLEX-P1-CARD-18-09-C16-BD1 Date: _____

Write what the two ANCA target, which immunofluorescence pattern each one gives, and the abbreviation used for each pattern. Then rebuild the mechanism in steps: what priming does to the neutrophil, where the target enzymes go, what the antibody does when it binds, what the neutrophil then releases, and which vessels are injured. Write what pauci-immune means and why these diseases are pauci-immune, and write what the biopsy of an immune complex vasculitis would show instead. Write which two organs are hit hardest and why those organs, naming the renal lesion with its urinary findings and the pulmonary lesion with its symptom. Write what a pulmonary-renal syndrome is and name the other disease that causes one, with the immunofluorescence finding that identifies it. Finish by building the three-column table of the ANCA-associated diseases from memory, with rows for ANCA type, granulomas, upper airway, lung, kidney, and the distinctive feature of each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C16-BD2

## Three patients who all cough blood

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C16 Activity: NPLEX-P1-CARD-18-09-C16-BD2 Date: _____

Patient 1 is a 58-year-old with two years of bloody nasal discharge and crusting, a collapsed nasal bridge, two cavitating lung nodules, hematuria with red cell casts, and PR3-ANCA. Patient 2 is a 63-year-old with hemoptysis, hematuria with red cell casts, palpable purpura on the legs, MPO-ANCA, and a renal biopsy showing crescents with almost no immunoglobulin. Patient 3 is a 41-year-old with twelve years of asthma and nasal polyps, a blood eosinophil count of 18%, patchy lung infiltrates, numbness and weakness in two separate nerve distributions, and new heart failure. For each patient name the disease and write the features that place them there rather than in either of the other two. Write what each renal biopsy would show and why the term pauci-immune applies. Explain why patient 1 has upper airway destruction and patient 2 does not. For patient 3 write the blood finding that drives the tissue injury, the proportion of such patients who are ANCA positive, and the organ involvement that is the main cause of death. Finish by writing what would make you suspect anti-GBM disease instead in a patient with lung hemorrhage and glomerulonephritis, and the single test finding that would show it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question  NPLEX-P1-CARD-18-09-C16-Q04

## Sinuses, cavities, and casts

**A 54-year-old woman has had 18 months of bloody nasal discharge with nasal crusting and a recently flattened nasal bridge. Chest imaging shows two cavitating nodules. Urinalysis shows hematuria with red cell casts, and the serum creatinine is 3.4 mg/dL. Renal biopsy shows necrotizing crescentic glomerulonephritis with almost no immunoglobulin or complement on immunofluorescence. Which of the following is the most likely serologic finding?**

A. Linear IgG deposition along the glomerular basement membrane

B. PR3-ANCA with a cytoplasmic immunofluorescence pattern

C. IgA-containing immune complexes in the glomerular mesangium

D. Hepatitis B surface antigen in circulating immune complexes

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C17-BD1

## IgA vasculitis and Behcet syndrome from mechanism to sign

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C17 Activity: NPLEX-P1-CARD-18-09-C17-BD1 Date: _____

Write what IgA vasculitis is, the age range, the usual preceding event, and the molecular abnormality that starts it, naming the modified antibody and what recognizes it. Write where the resulting complexes deposit, which complement pathways they activate, the hypersensitivity type, and what immunofluorescence shows. Then write the four organ manifestations, giving for each the mechanism and the typical finding, and write where the skin lesions sit and why that distribution. Write what the platelet count and coagulation tests show and which other cause of purpura that result excludes. Then write what Behcet syndrome is, the geographic distribution, the genetic association, and the cell type driving the inflammation. List its mucosal, eye, skin, vascular, and neurologic features, writing what pathergy is. Finish by writing the one finding in each disease that would make you choose it over the other.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C17-BD2

## Purpura in a child and ulcers in a young adult

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C17 Activity: NPLEX-P1-CARD-18-09-C17-BD2 Date: _____

A 6-year-old boy has raised purple spots that do not blanch over his buttocks and the backs of both legs, two weeks after a sore throat, with swollen painful knees and ankles, colicky abdominal pain, dark stools, and urine showing blood. His platelet count and coagulation tests are normal. Write the diagnosis, then explain each of his five findings mechanistically, one short paragraph each. Write why a normal platelet count is an important part of the picture and what it rules out. Write which abdominal complication he is at risk of and the usual segment involved, and write what a kidney biopsy would show and which separate renal disease it is identical to. Then a 26-year-old man of Turkish ancestry has recurrent painful mouth ulcers, genital ulcers, a painful red eye with a layer of pus in the anterior chamber, tender red shin nodules, and a calf deep vein thrombosis, and he develops a pustule where a needle was inserted. Write that diagnosis, the genetic association, the cell driving the inflammation, the name of the needle finding, and what the layer in the eye is called. Finish by writing why the term variable-vessel vasculitis is used for the second man and not the boy.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____